Think — inequalities

A Cambridge insert: when two quantities are not equal, and how to write and read that.

LESSON 20 | 1 × 40 MIN | MATEMATIKA 6, O‘YLAB KO‘R | S7 2.6

LESSON CLOCK

10' | 12' | 12' | 6'

Four symbols	10 min
On the number line	12 min
Listing the integers	12 min
In the world	6 min

LEARNING OBJECTIVES

- Read and write the four inequality symbols.
- Show an inequality on a number line.
- List the integers satisfying a simple inequality.
- Use inequalities to describe a real restriction.

TEXTBOOK

National	pp. 51–53
Cambridge	Stage 7, pp. 38–41
Workbook	Exercise 2.6

01 Explanation

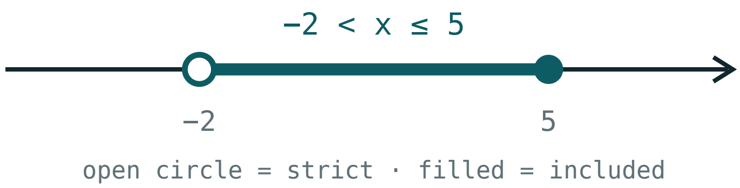
Four symbols

SYMBOL	READ AS	EVERYDAY PHRASE
>	is greater than	more than
<	is less than	fewer than
≥	is greater than or equal to	at least
≤	is less than or equal to	at most, no more than

THE WIDE END FACES THE LARGER NUMBER

$7 > 3$ and $3 < 7$ say the same thing. Reading the symbol aloud from left to right, every time, is how it stops being guessed at.

On the number line



An inequality shown as a ray on the number line

INEQUALITY	CIRCLE AT THE END	ARROW POINTS
$x > 3$	open	right
$x \geq 3$	filled	right
$x < 3$	open	left
$x \leq 3$	filled	left

> THE CIRCLE SAYS WHETHER THE END POINT COUNTS
>
> Filled means the number itself is included; open means it is not. That single dot is the whole difference between $>$ and $\geq$.

Listing the integers

CONDITION	THE INTEGERS	HOW MANY
$2 < x < 7$	$3, 4, 5, 6$	4
$2 \leq x \leq 7$	$2, 3, 4, 5, 6, 7$	6
$-3 < x \leq 1$	$-2, -1, 0, 1$	4
$x \geq 5$	$5, 6, 7, \ldots$	infinitely many

> $<$ EXCLUDES THE END, $\leq$ INCLUDES IT
>
> $2 < x < 7$ has four integers; $2 \leq x \leq 7$ has six. Two symbols, two extra numbers — and it is a mark in every paper that asks for a list.

In the world

STATEMENT	IN SYMBOLS
a lift carries at most 8 people	$n \le 8$
the class has at least 20 pupils	$n \ge 20$
the speed limit is $60\ km/h$	$v \le 60$
water is liquid between 0 and $100\ °C$	$0 < t < 100$
a triangle's angles	$0° < \alpha < 180°$

“AT MOST” IS ≤, NOT <

A lift that carries at most eight can carry eight. Everyday phrases translate into the symbols with the equality included far more often than pupils expect.

02 Worked examples

EXAMPLE 1 **List the integers satisfying $-3 < x \le 1$.**

1. -3 is excluded — the symbol is $<$.
2. 1 is included — the symbol is $\le$.
3. So: $-2, -1, 0, 1$.

ANSWER

$-2, -1, 0, 1$

EXAMPLE 2 **Write “a lift carries at most 8 people” in symbols.**

1. Let n be the number of people.
2. “At most” includes 8 itself.
3. $n \le 8$

ANSWER

$n \le 8$

EXAMPLE 3 **How many integers satisfy $2 \le x \le 7$?**

1. Both ends are included.
2. $2, 3, 4, 5, 6, 7$
3. Six integers.

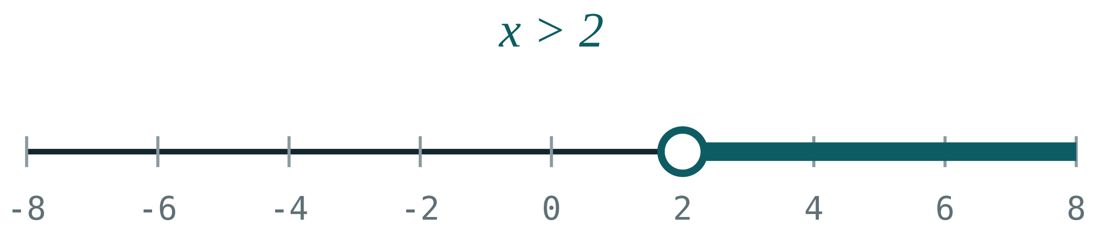
ANSWER

6

03 Interactive model

Ask for the number of pupils the room may hold and write it as an inequality on the board; the class supplies the ≤ themselves once they see that the full room is allowed.

An inequality on the number line



inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Inequality	Tengsizlik	Неравенство
Greater than	Katta	Больше
Less than	Kichik	Меньше
At least	Kamida	Не менее
At most	Ko‘pi bilan	Не более
Number line	Sonlar o‘qi	Числовая прямая
Solution set	Yechimlar to‘plami	Множество решений
Integer	Butun son	Целое число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\geq$ means:

- greater than
- at least
- at most
- less than

$\leq$ means:

- at least
- at most
- greater than
- not equal

An open circle on the line means the end point is:

- included
- excluded
- negative
- zero

The integers with $2 < x < 7$ number:

- 4
- 5
- 6
- 7

The integers with $2 \le x \le 7$ number:

- 4
- 5
- 6
- 7

“At most 8” is:

- $n < 8$
- $n \le 8$
- $n > 8$
- $n \ge 8$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Read $7 > 3$

   ANSWER Seven is greater than three

2. Read $3 < 7$

   ANSWER Three is less than seven

3. Read $x \geq 5$

   ANSWER x is at least five

4. Read $x \leq 5$

   ANSWER x is at most five

5. Is $-2 > -7$?

   ANSWER Yes

6. Is $0 < -1$?

   ANSWER No

7. The circle for $x > 3$

   ANSWER Open

Medium · the standard question

7 PROBLEMS

1. The integers with $2 < x < 7$

   ANSWER $3, 4, 5, 6$

2. The integers with $2 \le x \le 7$

   ANSWER $2, 3, 4, 5, 6, 7$

3. The integers with $-3 < x \le 1$

   ANSWER $-2, -1, 0, 1$

4. “A lift carries at most 8”

   ANSWER $n \le 8$

5. “The class has at least 20”

   ANSWER $n \ge 20$

6. “The speed limit is 60”

   ANSWER $v \le 60$

7. Water is liquid between 0 and 100 °C

   ANSWER $0 < t < 100$

Hard · stretch and reason

7 PROBLEMS

1. How many integers satisfy $-5 \le x < 4$?

   ANSWER 9

2. Write “more than 3 but no more than 10”

   ANSWER $3 < x \le 10$

3. The angles of a triangle satisfy

   ANSWER $0° < \alpha < 180°$

4. Which integers satisfy both $x > 2$ and $x \le 5$?

   ANSWER $3, 4, 5$

5. Can $x < 3$ and $x > 5$ both hold?

   ANSWER No

6. Write the ages of a Grade 6 pupil as an inequality

   ANSWER e.g. $11 \le a \le 13$

7. Why does “at most” use $\le$ and not $<$?

   ANSWER The stated number is itself allowed

07 Homework

Homework — 5 tasks

Read every symbol aloud from left to right before writing anything.

1. Write in symbols: at least 12; at most 30; more than 7.
2. List the integers satisfying $-4 \le x < 2$.
3. Show $x > -1$ on a number line, with the right kind of circle.
4. Write three real restrictions from your own day as inequalities.
5. How many integers satisfy $-10 < x \le 0$?